

Homework 3

ORF 245 - Fundamentals of Statistics

Due: Feb. 21st, 2025, End-of-Day (11:59pm)

Notes:

- Collaboration with other students is encouraged, but you must i) write up your own answers, and ii) acknowledge any sources or students you consulted.
- Submit the homework on Gradescope, and be sure to assign each page to the relevant question(s).

Exercise: 1. (Discrete Random Variables, PMF and CDF) Consider rolling a fair dice twice (independently), and let D be the random variable that equals the first roll minus the second roll. So, if the two dice come up $(1, 6)$ (i.e. the first die roll equals 1, and the second die roll equals 6) then $D = -5$; if the two dice come up $(6, 1)$ then $D = 5$.

(a) Find the probability mass function (pmf) of D .

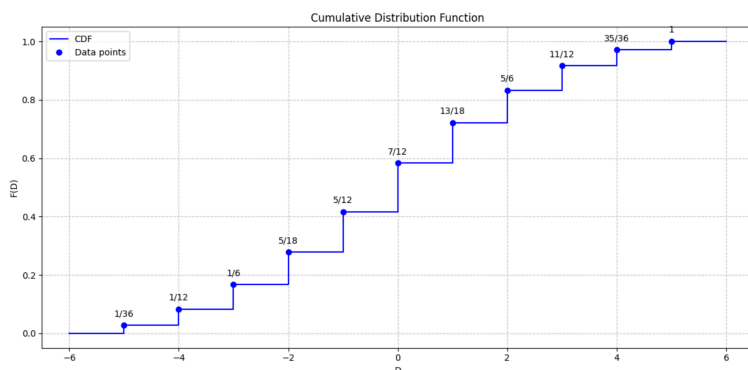
• Solution:

D	-5	-4	-3	-2	-1	0	1	2	3	4	5
$P(D)$	$\frac{1}{36}$	$\frac{1}{18}$	$\frac{1}{12}$	$\frac{1}{9}$	$\frac{5}{36}$	$\frac{1}{6}$	$\frac{5}{36}$	$\frac{1}{9}$	$\frac{1}{12}$	$\frac{1}{18}$	$\frac{1}{36}$

(b) Find and sketch the cumulative distribution function (cdf) of D .

• Solution:

D	-5	-4	-3	-2	-1	0	1	2	3	4	5
$F(D)$	$\frac{1}{36}$	$\frac{1}{12}$	$\frac{1}{6}$	$\frac{5}{18}$	$\frac{5}{12}$	$\frac{7}{12}$	$\frac{13}{18}$	$\frac{5}{6}$	$\frac{11}{12}$	$\frac{35}{36}$	1



- (c) Compute the expected value of D , $\mathbb{E}[D]$ and the expected value of the *absolute value* of D , $\mathbb{E}[|D|]$.

• **Solution:**

$$E[D] = \sum D \dot{P}(D) = 0, \mathbb{E}[|D|] = \sum |D| \dot{P}(D) = 35/18$$

- (d) Compute $\text{Var}(D)$

• **Solution:**

$$\text{Var}(D) = \mathbb{E}[D^2] - \mathbb{E}[D]^2 = \mathbb{E}[D^2] - 0 \approx 6.45$$

Exercise: 2. (Poisson Distribution) Let X have a Poisson distribution with parameter $\lambda > 0$. That is, the pmf of X is given by

$$p_X(k; \lambda) = \frac{\lambda^k e^{-\lambda}}{k!} \quad \text{for } k = 0, 1, \dots$$

For each part below, it will be helpful to use the series expansion for $e^\lambda = \sum_{k=0}^{\infty} \frac{\lambda^k}{k!}$.

- (a) Verify that this is a valid pmf (i.e., that it's non-negative and sums to 1).

• **Solution:**

For any $k \geq 0$ and $\lambda > 0$:

- λ^k is non-negative since any real number raised to a non-negative power is non-negative.
- $e^{-\lambda}$ is always positive for any real λ , as the exponential function is positive for all real inputs.
- $k!$ (factorial of k) is always positive for non-negative integers k .

Therefore, the product and division of these non-negative terms result in a non-negative value for $p_X(x; \lambda)$ for all k . Thus, the PMF is non-negative for all possible values of k .

To demonstrate the normalization, we need to show that the sum of $p_X(x; \lambda)$ over all possible values of k equals 1:

$$\sum_{k=0}^{\infty} p_X(k; \lambda) = \sum_{k=0}^{\infty} \frac{\lambda^k e^{-\lambda}}{k!} = 1$$

Using the series expansion for e^λ , which is given by $e^\lambda = \sum_{k=0}^{\infty} \frac{\lambda^k}{k!}$, we can rewrite the sum as:

$$\sum_{k=0}^{\infty} \frac{\lambda^k e^{-\lambda}}{k!} = e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{k!}$$

Notice that the series $\sum_{k=0}^{\infty} \frac{\lambda^k}{k!}$ is the expansion of e^λ . Therefore:

$$e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} = e^{-\lambda} e^\lambda = 1$$

Thus, the PMF sums to 1, satisfying the normalization condition and confirming that $p_X(x; \lambda)$ is a valid PMF for a Poisson distribution with parameter λ .

(b) Show that mean of X is $\mathbb{E}[X] = \lambda$.

• **Solution:**

Recall the definition of the expected value for discrete random variables:

$$\mathbb{E}[X] = \sum_{k=0}^{\infty} k \cdot p_X(k; \lambda)$$

Substituting the PMF of the Poisson distribution $p_X(k; \lambda) = \frac{\lambda^k e^{-\lambda}}{k!}$, we get:

$$\mathbb{E}[X] = \sum_{k=0}^{\infty} k \cdot \frac{\lambda^k e^{-\lambda}}{k!}$$

Simplifying the expression by taking $e^{-\lambda}$ outside the sum and cancelling one k with $k!$, we have:

$$\mathbb{E}[X] = e^{-\lambda} \sum_{k=1}^{\infty} \frac{\lambda^k}{(k-1)!}$$

By the change of variable $j = k - 1$, we rewrite the sum to start from $j = 0$:

$$\mathbb{E}[X] = e^{-\lambda} \sum_{j=0}^{\infty} \frac{\lambda^{j+1}}{j!} = \lambda e^{-\lambda} \sum_{j=0}^{\infty} \frac{\lambda^j}{j!}$$

Recognizing the series expansion of e^λ , the sum becomes:

$$\mathbb{E}[X] = \lambda e^{-\lambda} e^\lambda = \lambda$$

Thus, the mean of X , $\mathbb{E}[X]$, is equal to λ .

(c) Show that variance of X is $\text{Var}(X) = \lambda$.

(Hint: One way to approach this is to first compute $\mathbb{E}[X(X-1)]$ – why is this easier? How does it help?)

• **Solution:**

Using $E[X(X - 1)]$ simplifies the series expression by simplifying the factorials in it, as demonstrated below.

First, we compute $\mathbb{E}[X(X - 1)]$:

$$\mathbb{E}[X(X - 1)] = \sum_{k=0}^{\infty} k(k - 1) \cdot \frac{\lambda^k e^{-\lambda}}{k!}$$

Since the terms for $k = 0$ and $k = 1$ are zero, the sum effectively starts from $k = 2$:

$$\mathbb{E}[X(X - 1)] = e^{-\lambda} \sum_{k=2}^{\infty} \frac{\lambda^k}{(k - 2)!}$$

By the change of variable $j = k - 2$, we rewrite the sum to start from $j = 0$:

$$\mathbb{E}[X(X - 1)] = \lambda^2 e^{-\lambda} \sum_{j=0}^{\infty} \frac{\lambda^j}{j!}$$

Recognizing the series expansion of e^{λ} , the sum becomes:

$$\mathbb{E}[X(X - 1)] = \lambda^2 e^{-\lambda} e^{\lambda} = \lambda^2$$

Now, to find $\text{Var}(X)$, we use the fact that $\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$. We already know that $\mathbb{E}[X] = \lambda$ and can express $\mathbb{E}[X^2]$ as $\mathbb{E}[X(X - 1)] + \mathbb{E}[X]$ (Note that: $E[X^2] = E[X(X - 1) + X]$):

$$\text{Var}(X) = (\mathbb{E}[X(X - 1)] + \mathbb{E}[X]) - (\mathbb{E}[X])^2$$

Substituting the computed values:

$$\text{Var}(X) = (\lambda^2 + \lambda) - \lambda^2 = \lambda$$

Thus, the variance of X , $\text{Var}(X)$, is equal to λ .

Exercise: 3. (Validity of PDFs) Let

$$f(x) = \begin{cases} \frac{c}{x^2} & 1 \leq x \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

for some unknown constant c .

- (a) Find the value of c for which $f(\cdot)$ is a valid probability density function. Sketch or plot $f(\cdot)$.

- **Solution:**

By the definition of pdf, we must have

$$\int_{-\infty}^{\infty} f(x)dx = 1.$$

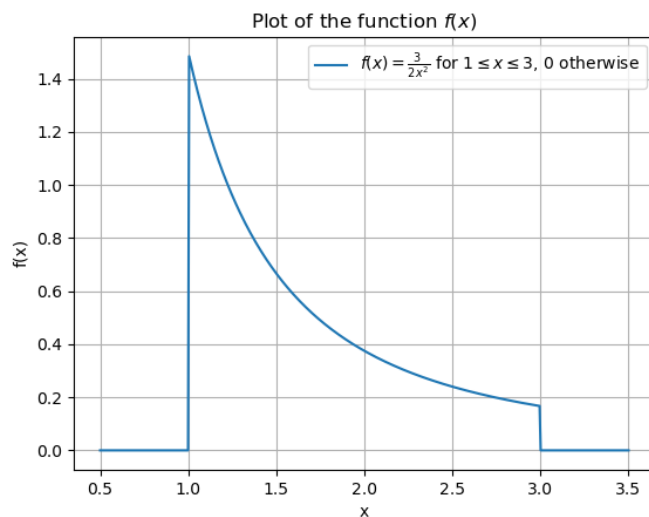
Plug in the expression of $f(x)$, we have

$$\begin{aligned} \int_1^3 \frac{c}{x^2} dx &= 1 \\ \iff \frac{2c}{3} &= 1 \\ \iff c &= \frac{3}{2}. \end{aligned}$$

Therefore

$$f(x) = \begin{cases} \frac{3}{2x^2} & 1 \leq x \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

The resulting plot of $f(x)$ is as the following.



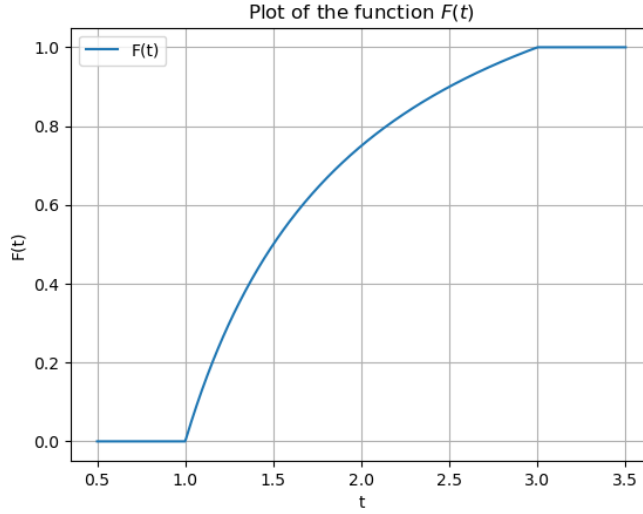
- (b) Let X be a continuous random variable with the pdf above. Find the corresponding cdf $F_X(x)$. Sketch or plot $F_X(\cdot)$.

- **Solution:**

By definition,

$$\begin{aligned} F_X(t) &= \int_{-\infty}^t f(x)dx \\ &= \begin{cases} 0 & t < 1 \\ \frac{3}{2}(1 - \frac{1}{t}) & 1 \leq t < 3 \\ 1 & t \geq 3 \end{cases} \end{aligned}$$

The plot of $F(t)$ is as the following.



(c) Compute $\mathbb{P}(1 \leq X \leq 2)$.

• **Solution:**

Since X is a continuous random variable,

$$\mathbb{P}(1 \leq X \leq 2) = F_X(2) - F_X(1) = \frac{3}{4}.$$

Exercise: 4. (A New Distribution?) Let X be a random variable with pdf

$$f(x) = \frac{1}{\sqrt{\pi}} \exp(-(x + \pi)^2)$$

(a) Find $\mathbb{E}[X]$ and $\text{Var}(X)$.

• **Solution:**

Notice that the pdf for a normal (Gaussian) random variable $\mathcal{N}(\mu, \sigma^2)$ is

$$f(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right).$$

Therefore by comparing $f(x)$ and $f(x; \mu, \sigma^2)$, we know that X follows the same distribution as $\mathcal{N}(-\pi, \frac{1}{2})$. Therefore $\mathbb{E}[X] = -\pi$, $\text{Var}(X) = \frac{1}{2}$.

(b) Find $\mathbb{E}[X^2]$.

• **Solution:**

Using the definition for variance of a random variable:

$$\mathbb{E}[X^2] = \text{Var}(X) + (\mathbb{E}[X])^2 = \frac{1}{2} + \pi^2.$$

(c) Let $Y = 2 - X/\pi$. Find $\mathbb{E}[Y]$ and $\text{Var}(Y)$.

• **Solution:**

$$\begin{aligned}\mathbb{E}[Y] &= \mathbb{E}[2 - X/\pi] \\ &= 2 - \mathbb{E}[X]/\pi \\ &= 3.\end{aligned}$$

$$\begin{aligned}\text{Var}(Y) &= \text{Var}(2 - X/\pi) \\ &= \text{Var}(-X/\pi) \\ &= \left(-\frac{1}{\pi}\right)^2 \text{Var}(X) \\ &= \frac{1}{2\pi^2}.\end{aligned}$$

(d) Find $\mathbb{P}(X > \pi)$ in terms of $\Phi(z)$, the cdf of a standard normal random variable.

• **Solution:**

Notice that $X \sim \mathcal{N}(-\pi, \frac{1}{2})$ implies $X + \pi \sim \mathcal{N}(0, \frac{1}{2})$, and further implies $\sqrt{2}(X + \pi) \sim \mathcal{N}(0, 1)$. Therefore

$$\begin{aligned}\mathbb{P}(X > \pi) &= \mathbb{P}(\sqrt{2}(X + \pi) > \sqrt{2}(\pi + \pi)) \\ &= \mathbb{P}(\sqrt{2}(X + \pi) > 2\sqrt{2}\pi) \\ &= 1 - \mathbb{P}(\sqrt{2}(X + \pi) \leq 2\sqrt{2}\pi) \\ &= 1 - \Phi(2\sqrt{2}\pi).\end{aligned}$$

Exercise: 5. [OPTIONAL] (Conditional Geometric Distribution)

Note: This is an *optional* exercise – it will not be graded, will not factor into your grade (either positively or negatively), but we will release solutions after the submission deadline.

Recall that the probability mass function of a geometric random variable (parameterized by α) is

$$p_X(x; \alpha) = \begin{cases} (1 - \alpha)^{x-1} \alpha & \text{for } x = 1, 2, \dots \\ 0 & \text{otherwise} \end{cases}$$

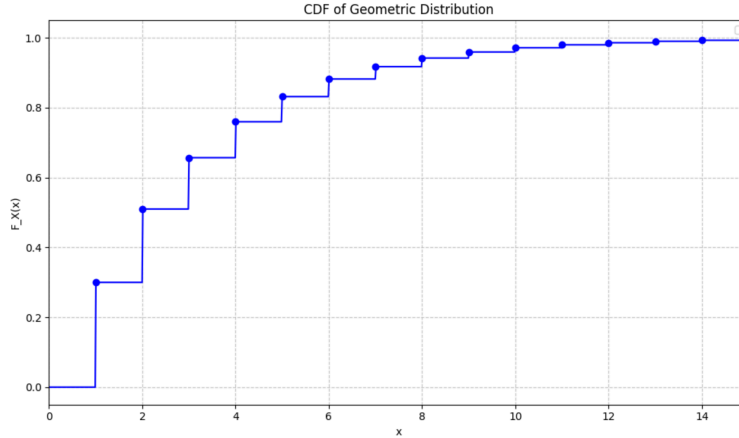
Suppose $X \sim \text{Geo}(\alpha)$, i.e., the pmf of X is as given above, for some $\alpha \in (0, 1]$.

(a) Find and sketch the cdf of X . Make sure to label your axes.

• **Solution:**

$$F(x) = 1 - (1 - \alpha)^{\lfloor x \rfloor} \text{ for } x \geq 1 \quad (1)$$

$$F(x) = 0 \text{ for } x < 1 \quad (2)$$



(b) For a fixed $t \geq 1$, find $\mathbb{P}(X \geq t)$.

• **Solution:**

$$\mathbb{P}(X \geq t) = (1 - \alpha)^{\lceil t \rceil - 1}$$

(c) For a fixed $t \geq 1$, compute $\mathbb{P}(X = x | X \geq t)$ for $x \geq t$. I.e. compute the probability that X equals x conditioned on the event $X \geq t$.

This is called the *conditional pmf of X given $X \geq t$* , and is denoted $p_{X|X \geq t}(x; \alpha)$.

• **Solution:**

$$\text{For } t \geq 1, p_{X|X \geq t}(x; \alpha) = \begin{cases} \alpha(1 - \alpha)^{x - \lceil t \rceil} & \text{for } x = \lceil t \rceil, \lceil t \rceil + 1, \dots \\ 0 & \text{otherwise} \end{cases}$$

(d) For some event A and discrete random variable X , define the *conditional expectation of X given A* as $\mathbb{E}[X|A] = \sum_{x \in \text{range}(X)} x \cdot p_{X|A}(x; \alpha)$, where $p_{X|A}(x)$ is the conditional probability mass function of X given event A .

Compute $\mathbb{E}[X|X \geq t]$.

(Hint: you can use the fact that $\mathbb{E}[X] = \frac{1}{\alpha}$.)

• **Solution:**
$$\mathbb{E}[X | X \geq t] = \sum_{x \in \lceil t \rceil, \lceil t \rceil + 1, \dots} x \alpha (1 - \alpha)^{x - \lceil t \rceil} = \lceil t \rceil - 1 + \sum_{x \in 1, 2, \dots} x \alpha (1 - \alpha)^{x - 1} = \lceil t \rceil + \frac{1 - \alpha}{\alpha}$$

(e) Provide a one-sentence interpretation of the results in part (d) – why might the geometric distribution be called “memoryless”?

(Hint: recall that geometric random variables are often used to model a “waiting time”.)

- **Solution:**

Regardless of how long you've already waited (t), the expected additional waiting time $(1 - \alpha)/\alpha$ remains constant - as if the distribution has no "memory" of the time that has already passed.